

034IN - FONDAMENTI DI AUTOMATICA -
FUNDAMENTALS OF AUTOMATIC CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION
A.Y. 2023/2024
SOLUTION

January 30, 2025

First and Last Name:

Group: Group A

Exercise: Exercise 1 - 2 Questions

Available Time Frame: 60 minutes

Important Notes:

- Write your answers on a single white sheet of paper using a black pen.
- Do not write in blue ink or pencil.
- Do not hand in additional sheets.
- The clarity and precision of your answers will be subject to evaluation.

I declare that the answers to this exercise are my own and only my own work and that I have not consulted with others.

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ADDITIONAL PARTIAL WRITTEN EXAMINATION – A.Y. 2023/2024
January 30, 2025

First and Last Name:

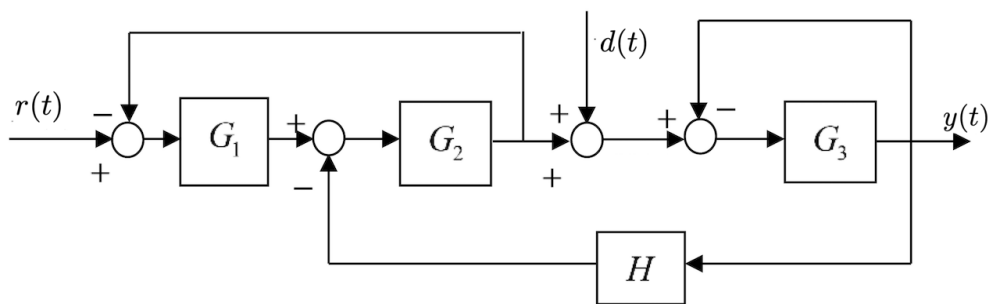
Group: Group A

Exercise: Exercise 1

Available Time Frame: 60 minutes

Exercise 1

The closed-loop system shown in the block diagram has two input signals, namely $r(t)$ and $d(t)$, and one outputs, $y(t)$.



Question 1.1.

- Find the transfer functions $T_r^y(s)$, from the input $r(t)$ to the output $y(t)$, and $T_d^y(s)$ from the input $d(t)$ to the output $y(t)$.

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ADDITIONAL PARTIAL WRITTEN EXAMINATION – A.Y. 2023/2024
September 16, 2024

First and Last Name:

Group: Group A

Exercise: Exercise 2

Available Time Frame: 60 minutes

Exercise 2

Consider an LTI system with the following characteristic polynomial:

$$p(s) = 3s^3 + (2K - 1)s^2 + 12s + (8K - 4)$$

where $K \in \mathbb{R}$.

Question 2.1.

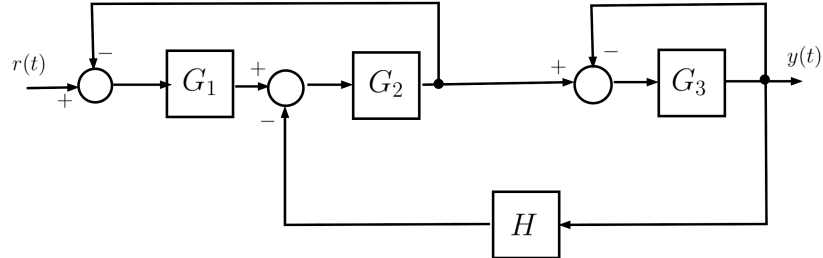
- Analyse the system's stability as a function of the parameter K .
- What are the roots of the polynomial $p(s)$, when $K = 0$?
- Is there a value of K such that at least one of the roots of the polynomial is $\bar{s} = -2$?
- Is there a value of K such that at least one of the roots of the polynomial is $\bar{s} = -4$?

Comment on your findings.

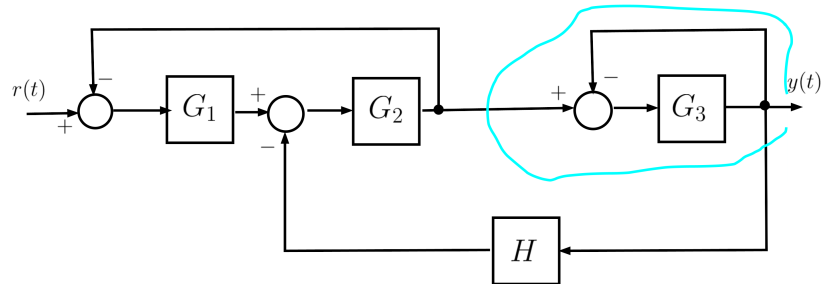
EXERCISE 1 – A SOLUTION PROPOSAL

Let's determine the transfer functions $T_r^y(s)$ from the input $r(t)$ to the output $y(t)$:

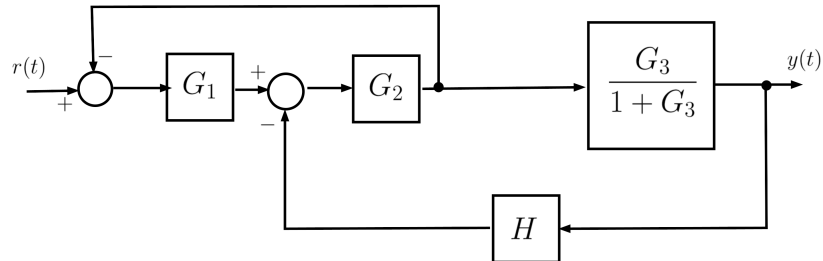
- Let's redraw the block diagram, putting the second input $d(t)$ to zero



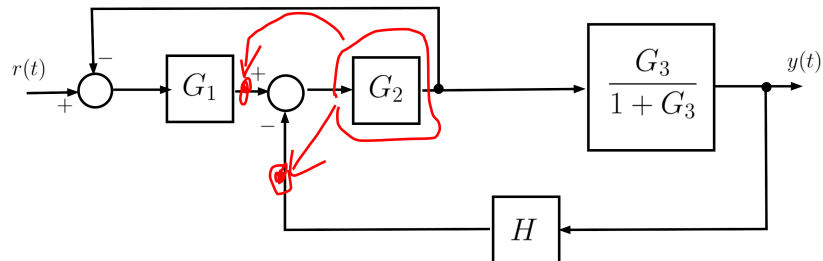
- Simplify the part of the diagram highlighted in cyan



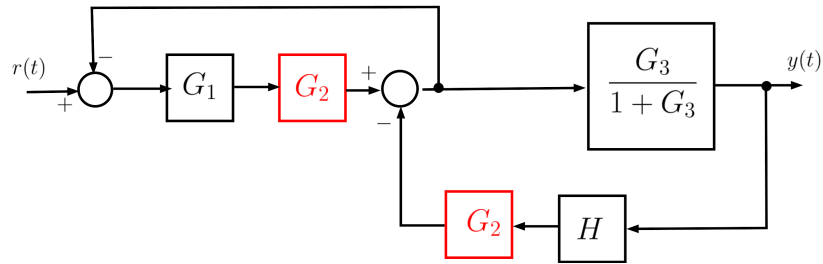
obtaining the following diagram



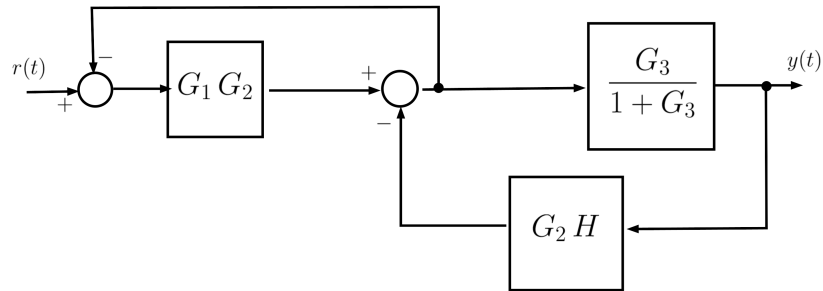
- Move the block with the transfer function G_2 upstream of the summing node



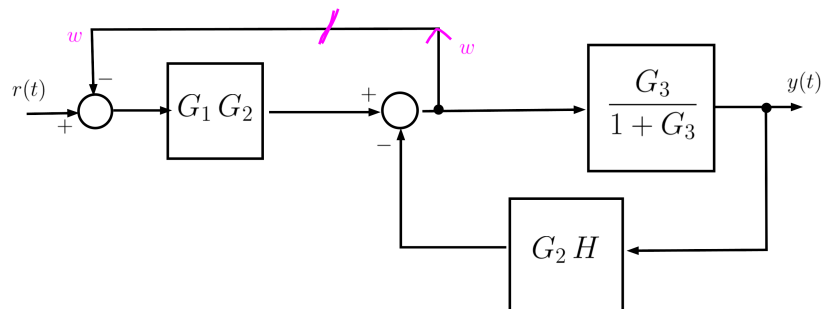
- The block diagram now results



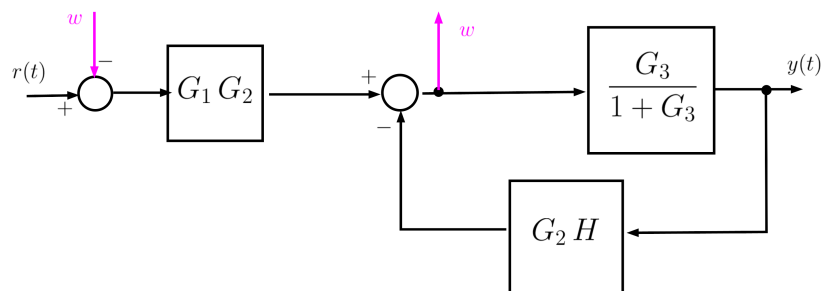
- Simplify the blocks connected in series:



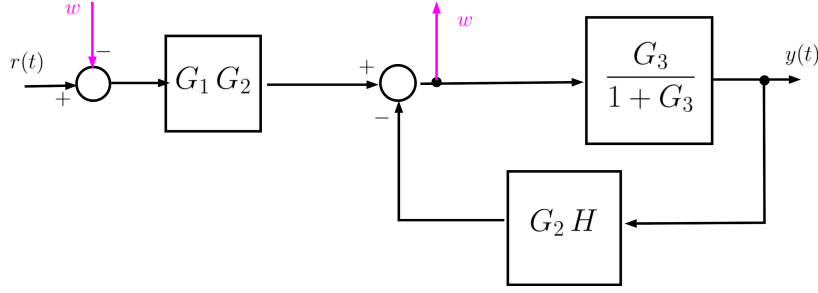
- It is now convenient to cut the highlighted arc in magenta and then use the **linear superposition theorem**. Suppose more than one input affects an output (in our case, two inputs, namely r and w , affect each output, i.e., y and w).



In that case, the effects of all inputs on every output can be determined simply as the algebraic sum of the effect (i.e., the transfer function) of each input taken separately on the considered output.



- Consider the diagram



The effects of the inputs r and w on the output y can be determined as

$$Y(s) = T_r^y R(s) + T_w^y W(s)$$

where $Y(s)$, $R(s)$ and $W(s)$ are the Laplace transforms of $y(t)$, $r(t)$ and $w(t)$ respectively. The terms T_r^y and T_w^y are, respectively, the transfer function from the input r (taken separately) to the output y and the transfer function from the input w (taken separately) to the output y . Similarly, the effects of the inputs r and w on the output w can be determined as

$$W(s) = T_r^w R(s) + T_w^w W(s)$$

Thus, we obtain

$$Y = \left[G_1 G_2 \frac{\frac{G_3}{1+G_3}}{1 + \frac{G_2 G_3 H}{1+G_3}} \right] \cdot (R - W)$$

$$W = \left[G_1 G_2 \frac{1}{1 + \frac{G_2 H G_3}{1+G_3}} \right] \cdot (R - W)$$

Rearranging the last expression we obtain:

$$\left[1 + \frac{G_1 G_2 (1+G_3)}{1+G_3+G_2 G_3 H} \right] W = \frac{G_1 G_2 (1+G_3)}{1+G_3+G_2 G_3 H} R \Rightarrow W = \frac{G_1 G_2 (1+G_3)}{(1+G_3)(1+G_1 G_2)+G_2 G_3 H} R$$

- Finally, by substituting the expression of W into that of Y we obtain the first requested transfer function:

$$Y = \left[G_1 G_2 \frac{\frac{G_3}{1+G_3}}{1 + \frac{G_2 G_3 H}{1+G_3}} \right] \cdot \left[1 - \frac{G_1 G_2 (1+G_3)}{(1+G_3)(1+G_1 G_2)+G_2 G_3 H} \right] R$$

$$= \frac{G_1 G_2 G_3}{1+G_3+G_2 G_3 H} \cdot \frac{1+G_3+G_1 G_2+G_1 G_2 G_3+G_2 G_3 H-G_1 G_2-G_1 G_2 G_3}{(1+G_3)(1+G_1 G_2)+G_2 G_3 H} R$$

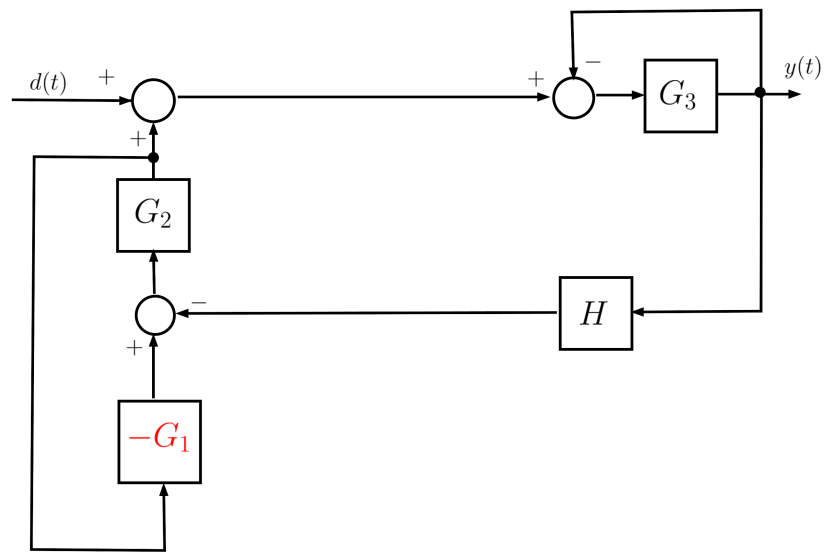
$$= \frac{G_1 G_2 G_3}{(1+G_3)(1+G_1 G_2)+G_2 G_3 H} R$$

Thus

$$T_r^y = \frac{G_1 G_2 G_3}{(1+G_3)(1+G_1 G_2)+G_2 G_3 H}$$

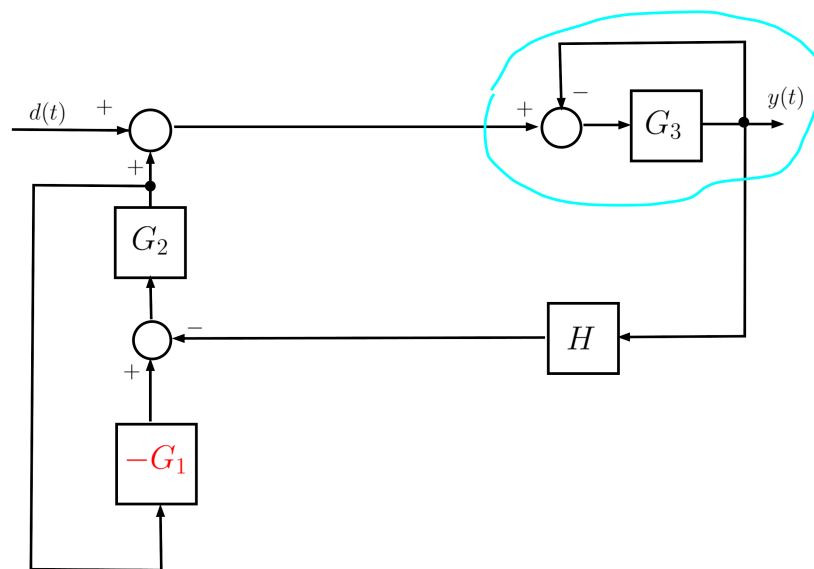
Let's determine the transfer functions $T_d^y(s)$ from the input $d(t)$ to the output $y(t)$:

- Let's redraw the block diagram, putting the first input $r(t)$ to zero and :

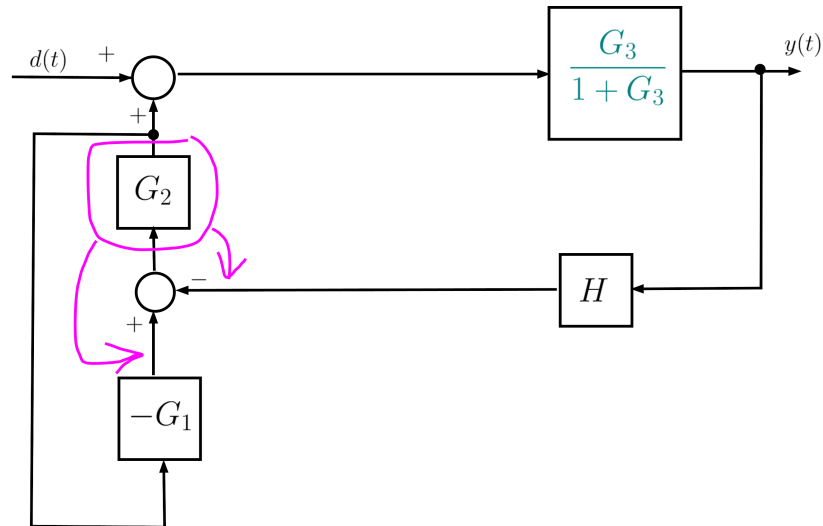


Note the sign of the term $-G_1$.

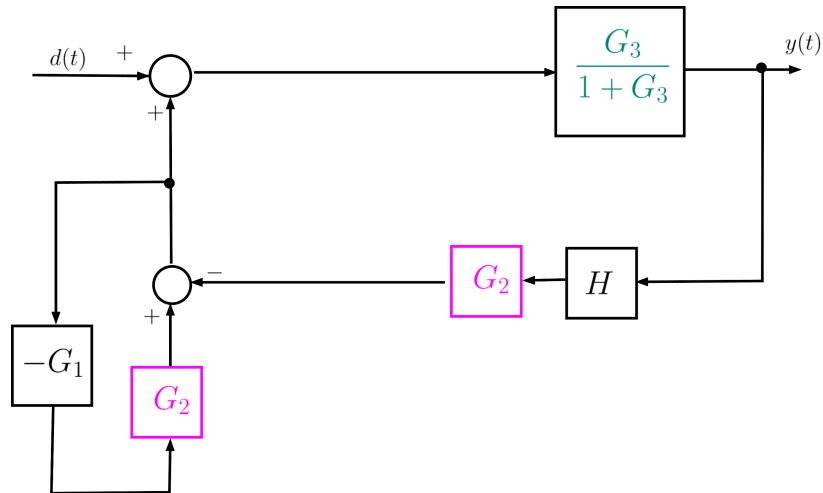
- Simplify the portion of the block diagram highlighted in cyan



- Move the block G_2 upstream of the summing node

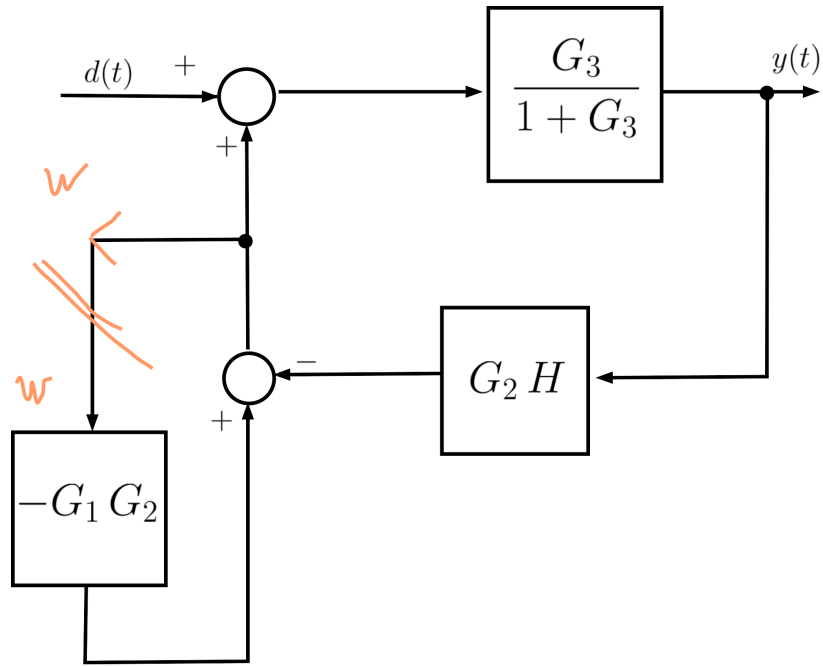


obtaining the following block diagram. Let's simplify the block connected in series

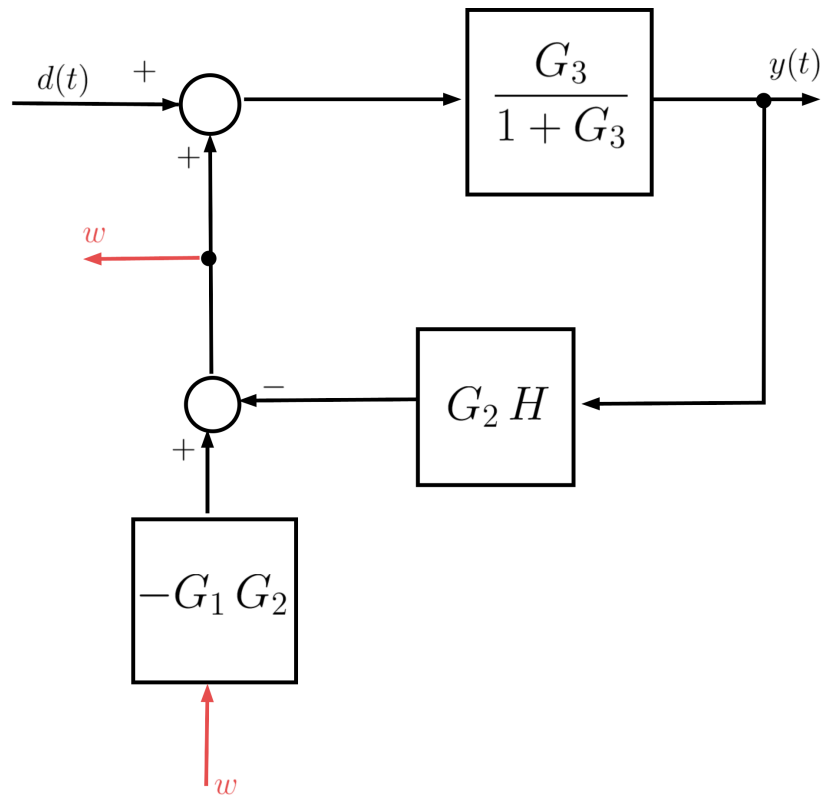


- Now it is convenient (once more) to cut the highlighted arc in orange and the use the linear superposition theorem. Two inputs, namely r and w , affect each output, i.e., y and w . In that case, the effects of all inputs on every output can be determined simply as the algebraic sum of the effect (i.e., the transfer function) of each input taken separately on the considered output.

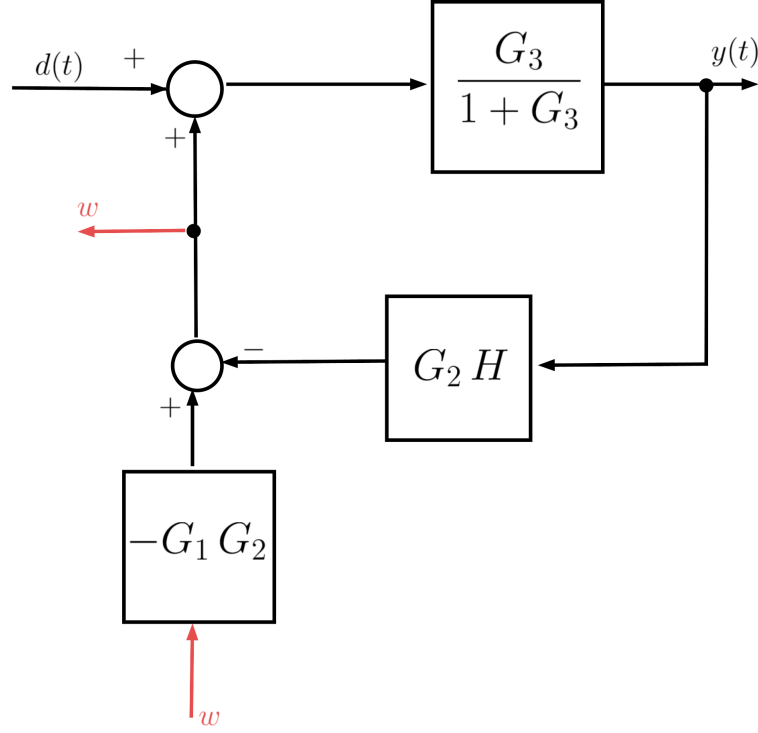
- Before the cut, we have



- Once the cut has been made, the diagram looks like this



- Consider the diagram



The effects of the inputs d and w on the output y can be determined as

$$Y(s) = T_d^y D(s) + T_w^y W(s)$$

where $Y(s)$, $D(s)$ and $W(s)$ are the Laplace transforms of $y(t)$, $d(t)$ and $w(t)$ respectively. The terms T_d^y and T_w^y are, respectively, the transfer function from the input d (taken separately) to the output y and the transfer function from the input w (taken separately) to the output y . Similarly, the effects of the inputs d and w on the output w can be determined as

$$W(s) = T_d^w D(s) + T_w^w W(s)$$

Thus, we obtain

$$\begin{aligned} Y &= \left[\frac{G_3}{1 + G_3 + G_2 G_3 H} \right] \cdot D + \left[-\frac{G_1 G_2 G_3}{1 + G_3 + G_2 G_3 H} \right] \cdot W \\ W &= \left[-\frac{G_2 G_3 H}{1 + G_3 + G_2 G_3 H} \right] \cdot D + \left[-G_1 G_2 \frac{1 + G_3}{1 + G_3 + G_2 G_3 H} \right] \cdot W \end{aligned}$$

Rearranging the last expression we obtain:

$$\left[1 + G_1 G_2 \frac{1 + G_3}{1 + G_3 + G_2 G_3 H} \right] W = \left[-\frac{G_2 G_3 H}{1 + G_3 + G_2 G_3 H} \right] \cdot D \Rightarrow W = -\frac{G_2 G_3 H}{(1 + G_3) (1 + G_1 G_2) + G_2 G_3 H} D$$

- Finally, by substituting the expression of W into that of Y we obtain the first requested transfer function:

$$\begin{aligned} Y &= \left[\frac{G_3}{1 + G_3 + G_2 G_3 H} \right] \cdot \left[1 + G_1 G_2 \cdot \frac{G_2 G_3 H}{(1 + G_3) (1 + G_1 G_2) + G_2 G_3 H} \right] \cdot D \\ &= \frac{G_3 (1 + G_1 G_2) (1 + G_3 + G_2 G_3 H)}{(1 + G_3 + G_2 G_3 H) [(1 + G_3) (1 + G_1 G_2) + G_2 G_3 H]} D \\ &= \frac{G_3 (1 + G_1 G_2)}{(1 + G_3) (1 + G_1 G_2) + G_2 G_3 H} D \end{aligned}$$

Thus

$$T_d^y = \frac{G_3 (1 + G_1 G_2)}{(1 + G_3) (1 + G_1 G_2) + G_2 G_3 H}$$

Concluding, the requested transfer functions are:

$$T_r^y = \frac{G_1 G_2 G_3}{(1 + G_3) (1 + G_1 G_2) + G_2 G_3 H}$$

$$T_d^y = \frac{G_3 (1 + G_1 G_2)}{(1 + G_3) (1 + G_1 G_2) + G_2 G_3 H}$$

EXERCISE 2 – A SOLUTION PROPOSAL

- The analysis of the system's stability as a function of the parameter K .

Consider the polynomial

$$p(s) = 3s^3 + (2K - 1)s^2 + 12s + 4(2K - 1)$$

and let's determine the terms into the Routh table

$$\begin{array}{c|cc} 3 & 3 & 12 \\ 2 & 2K - 1 & 4(2K - 1) \\ 1 & 0 & \\ 0 & & \end{array}$$

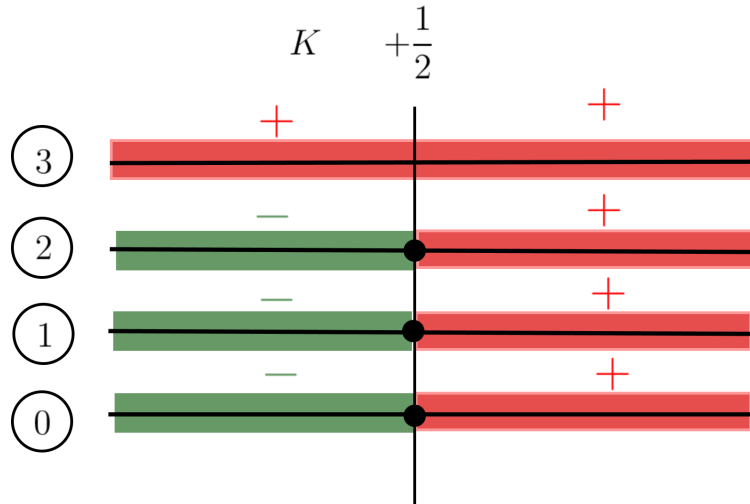
Note: the whole table's row 1 is null: we can substitute the null row with the coefficients of the first derivative of the **auxiliary polynomial** $q(s)$ obtained using the coefficients of the row immediately above the null one:

$$\begin{array}{c|cc} 3 & 3 & 12 \\ 2 & 2K - 1 & 4(2K - 1) \\ 1 & 0 & \\ 0 & & \end{array} \implies q(s) = (2K - 1)(s^2 + 4) \implies \frac{dq(s)}{ds} = 2(2K - 1)s$$

Thus, we can complete the Routh table as follows:

$$\begin{array}{c|cc} 3 & 3 & 12 \\ 2 & 2K - 1 & 4(2K - 1) \\ 1 & 2(2K - 1) & \\ 0 & 4(2K - 1) & \end{array}$$

Let's analyse the sign of the terms in the first column of the Routh's table:



In conclusion:

- $K < \frac{1}{2}$: you find one sign variation followed by one sign permanency when considering the sign of the terms into the first column of the Routh's table: the **system** is **unstable** (due to the sign variation);

- When $K = \frac{1}{2}$ the polynomial $p(s)$ becomes

$$p(s) = 3s^3 + 12s = 3s(s^2 + 4)$$

So the polynomial roots (the system's eigenvalues) are

$$\lambda_1 = 0, \quad \lambda_{2,3} = \pm j2$$

The roots have null real part and they have unitary algebraic multiplicity: the system is **marginally stable**.

- Finally, for $K > \frac{1}{2}$ there is no sign variations when considering the sign of the terms into the first column of the Routh's table. Keep in mind that you found a so-called "critical case" during the evaluation of the terms in the Routh's table (i.e. the null row with index "1"). Thus there is **no asymptotic stability in this case**. Indeed, the polynomial $q(s) = s^2 + 4$ is a factor of $p(s)$

$$p(s) = (s^2 + 4) \cdot (3s + 2K - 1)$$

We can conclude that also in this case the polynomial $p(s)$ corresponds to a **marginally stable system**.

- What are the roots of the polynomial $p(s)$, when $K = 0$?

Simply substitute the value 0 for the parameter K in the $p(s)$ expression, getting:

$$\begin{aligned} K = 0 \implies p(s) = 3s^3 - s^2 + 12s - 4 = (s^2 + 4) \cdot (3s - 1) \implies \\ \lambda_1 = +\frac{1}{3} \\ \lambda_{2,3} = \pm j2 \end{aligned}$$

The eigenvalue λ_1 is a real, positive value: the **system is unstable**.

- Is there a value of K such that at least one of the roots of the polynomial is $\bar{s} = -2$?

Evaluate the polynomial $p(s)$ for $s = -2$:

$$p(-2) = 3 \cdot (-8) + (2K - 1)(+4) + 12(-2) + 8K - 4 = -48 + 8(2K - 1) = 0 \implies K = \frac{7}{2}$$

- Is there a value of K such that at least one of the roots of the polynomial is $\bar{s} = -4$?

Evaluate the polynomial $p(s)$ for $s = -4$:

$$p(-4) = 3 \cdot (-64) + (2K - 1)(+16) + 12(-4) + 8K - 4 = -260 + 40K = 0 \implies K = \frac{13}{2}$$